

The Two-Strategy Playbook

Weather Trading + Sports Value-Betting

Dense Edition — every line is signal

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Educational only — not financial advice. Prediction-market trading is risky, zero-sum, and legally restricted in many places including Australia. Never trade money you cannot afford to lose.

The Two-Strategy Playbook

Weather Trading + Sports Value-Betting on Prediction Markets — Dense Edition

Information-dense. Every line is signal. Read top to bottom once, then use as a reference.

THE ONE SKILL (both strategies are this)

Buy when the market price is below the true probability — measured against a reliable "answer key" — then hold to resolution.

- Weather's answer key = **free professional forecast models**.

- Sports' answer key = **de-vigged sharp bookmaker odds (Pinnacle)**.

Everything below is how to (a) get a trustworthy true probability, (b) find where Polymarket disagrees, (c) size it, (d) prove your edge is real.

BRUTAL TRUTHS (memorize)

- **~88% of users lose** (only ~12% net-positive; only ~3% are "skilled" at value betting). The edge is real but thin and statistical.

- **Never bet money you can't afford to lose. Never your whole bankroll. Not even close.** A normal 6-loss streak wipes out a big-bet account even with a *winning* system ("risk of ruin").

- **Access:** Polymarket **and** Kalshi both block **Australia** (ACMA ISP-block since Aug 2025). A VPN breaks ToS and gives zero fund-recovery protection. Solve "where can I legally trade" *before* funding. The skill below is portable; the access problem is separate.

- **Costs are real:** ~2% **winner fee** + **spread** (1–2¢ liquid, 5–10¢ niche). A 5% edge in a 6% spread is a *loser*. Always subtract costs first.

- **Validate before scaling.** Most strategies "fail" not because the logic is wrong but because the edge was never proven. Paper-trade, log everything, judge after 100+ trades.

PART A — SHARED FOUNDATIONS

A1. Price = probability

A share pays **\$1 if right, \$0 if wrong**. Price in cents = the crowd's implied probability (65¢ = 65%). "Yes" + "No" ≈ \$1.

A2. Expected Value (EV) — the only question that matters

EV = (your true prob × amount won) – (lose prob × amount lost). Positive EV = profitable on average → bet. Negative = skip, no matter how tempting.

Example: buy "Yes" at 40¢, you think true 65%. Win → +60¢ (65%); lose → –40¢ (35%). $EV = 0.65 \times 0.60 - 0.35 \times 0.40 = +\$0.25/\text{share}$. Then subtract costs — if spread+fee = 6¢, real EV ≈ +19¢. Still a great bet.

A3. Costs (and how to dodge them)

- **Taker fee** ~2% (weather ~1.25%), **highest near 50¢**, ~0 near 1¢/99¢.
- **Maker = 0 fee**. Post a **limit order** (offer to buy at your price, wait) instead of a market order → pay no fee. Use limit orders by default.
- **Spread** = hidden round-trip cost. Worst in niche markets (where the best edges live) → demand a bigger edge there.

A4. Sizing — fractional Kelly

- Kelly fraction: $f = (b \cdot p - q) / b$ — b = decimal odds – 1, p = your true prob, $q = 1 - p$. f = % of bankroll.
- **Use Quarter Kelly (0.25×)** — full Kelly assumes you *know* the true prob (you don't); fractional cuts variance and protects against estimate error.
- **Hard caps:** ≤5% of bankroll per position; while learning, a flat tiny stake (\$1–5).
- **Bankroll** = money you can fully afford to lose. Diversify across many small bets — the edge only shows over *hundreds* of independent trades.

A5. Proof of edge (do this or you're guessing)

- **Log every trade:** date, market, your prob, price paid, size, result.
 - For sports, also log **Closing Line Value (CLV)** — see C5. CLV proves edge *before* results catch up.
 - Judge after **100+ trades:** are you positive *after costs*? If not, fix it before risking more.
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PART B — WEATHER TRADING

B1. Why it's good

Objective government resolution (no rug-pulls), casual/emotional opponents, **free pro forecasts** they ignore, **daily** feedback (fast learning). The big winners: gopfan2 (~\$2M), ColdMath (~\$120k+ barbell). An honest public bot (natestokens) logged **+27% ROI, 61% win** — real, modest, repeatable.

B2. How the markets work

- Question split into **temperature buckets** (e.g. ≤ 70 , 71–72, 73–74, 75–76, 77+), each a Yes/No ticket; prices sum to ~100%.
- **Resolves on the official NWS Daily Climate Report (CLI) for ONE specific airport station**, listed in the market rules. Settles **8:00 AM ET next day**. ~212 weather markets; cities incl. NYC, London, Guangzhou, Beijing, Shanghai, Miami, Hong Kong, + secondary cities.
- ■■ **The #1 beginner error**: analyzing the city, not the **airport**. NYC resolves at **LaGuardia (KLGA)** — can run **3–5°F cooler** than Midtown. Dallas = **Love Field (KDAL)**, not DFW. Always analyze the exact station.

B3. The data (all free)

- **Models**: GFS (US), ECMWF (Europe, sharpest), ICON (Germany), HRRR (US short-range).
- **Model runs (your timing edge)**: GFS 00/06/12/18 UTC; ECMWF 00/12 UTC. When a new run shifts the forecast, the *price lags* → trade before it catches up.
- **Ensembles = probability**: run the model many times; if 25 of 31 members say "75°F+", that's ~80%. Best bots use 30–170+ members.
- **Sources**: **Open-Meteo** (free ensembles incl. 31-member GEFS & ECMWF), **api.weather.gov** (NWS forecasts + live **METAR** airport obs), **Windy** ("compare models" GFS/ECMWF/ICON), **Tropical Tidbits** (raw model output).

B4. The edge & entry checklist

Edge = **your ensemble probability** – **market price**. Buy a bucket only if ALL hold:

1. Edge $\geq$ **8 percentage points** (asymmetric ok: ~5pt to fade an overpriced bucket "No", ~15pt to back "Yes").
2. Lead time $\leq$ **48 hours** (forecasts reliable).
3. **GFS & ECMWF ensembles agree within ~1.5°C** (if models fight, sit out).
4. Market volume $\geq$ **~\$10k** (so you can enter/exit).
5. **Avoid the coin-flip middle** — leave a **3°F buffer** around the ensemble mean; trade buckets the models clearly favor/reject.

If one fails → **don't trade**. Skipping is a profitable action.

B5. Advanced intraday edge — advection / nowcasting

Advection = a warm/cold air mass blowing in. The day-ahead forecast guessed; the **live METAR** airport readings show *reality*. If warm air is arriving faster than forecast, the high will likely beat it → trade before

the market notices. "Nowcasting" = using current obs to refine the forecast. (Level-2 skill; free via api.weather.gov.)

B6. Cities, tactics, sizing

- **Trade SECONDARY cities** (Buenos Aires, Cape Town, Dallas, Atlanta): fewer bots, edges last **hours** not 5–15 min. Avoid bot-saturated NYC/London/Hong Kong (edge gone in minutes) — but they have tighter spreads. Trade-off: secondary = bigger edge but wider spread (clear it).
- **Laddering**: buy several adjacent cheap buckets (5–15¢); one or two winners cover the rest.
- **Barbell (ColdMath)**: many tiny long-shot buckets + occasional bigger high-conviction bets.
- **Sizing**: learning = flat \$1–5; proven = Quarter Kelly, $\leq 5\%$ bankroll, run **10–30+ city/date combos** small (variance demands volume).
- **Costs**: weather taker fee $\sim 1.25\%$; use **limit orders (0 fee)**; ensure the ≥ 8 pt edge clears fee + spread.

B7. Automate it (you have Claude Code)

The bot = the manual routine, tireless, across 30 cities. Best open-source blueprints:

- **natestokens/polymarket-weather-bot** (Claude-Code-built, the gold standard): **173-member ensemble** (ECMWF IFS+AIFS, GEFS, ICON via Open-Meteo); **city bias-correction** vs historical ASOS airport data (NYC +1.4°F, Miami –2.2°F); edge = model prob – price, thresholds **5pt (No) / 15pt (Yes)**, "stay out of the middle"; pure Python, no deps; **verified +27% ROI / 61% win**.
 - **alteregoeth-ai/weatherbot**: ECMWF+HRRR via Open-Meteo, METAR, Visual Crossing, Gamma API; config min EV 0.05, Kelly 0.25, max bet \$20, scan hourly; has **simulation/paper mode**.
 - Official plumbing: **Polymarket/polymarket-cli** (Rust, JSON for agents) or **py-clob-client** (Python).
- Paper-trade for weeks first.**

B8. Manual routine (no code) & realistic expectations

Daily: pick city → find its airport station → check Windy compare-models for tomorrow's high → if 3+ models cluster on a range (true prob 70–90%) → open the market → find a bucket the models favor selling cheap (or one they reject selling rich) → run the B4 checklist → place a small **limit** order → **log it** → repeat → review after 100 trades.

Expect: a small grinding edge ($\sim 25\%$ ROI / $\sim 60\%$ win at small scale), not a jackpot. Losing days/weeks are normal at 60% win. The "\$80k/month" videos are marketing.

Common killer: "had a theory, won a few weeks, called it an edge, then lost." Validate with logged data, not vibes.

PART C — SPORTS VALUE-BETTING

C1. Why it's the most logically bulletproof

Pinnacle's closing line $\approx$ true probability. Hard evidence: across **397,935 football games**, $r^2 = 0.997$ between Pinnacle's closing line and actual outcomes; optimal prediction puts **95–100% weight on Pinnacle** over all other books. Why: Pinnacle's **"winner's welcome"** policy invites sharp bettors and uses their money to price lines (other books *ban* winners, so their lines are softer). **Polymarket sports prices are set by slow retail money.** The gap between de-vigged Pinnacle and Polymarket = your edge. Documented human: **"swisstony" ~\$3.7M** doing exactly this. The premise is the hardest in this whole field to argue with — the only real flaws are *execution and access*, both of which you control.

C2. The full mechanic

1. Get the **sharp line** (Pinnacle, or Betfair Exchange / PS3838) for the game.
2. **De-vig** it $\rightarrow$ true probability (C3).
3. Compare to the **Polymarket price**.
4. If Polymarket price < true prob by $\geq$ **your threshold (after costs)** $\rightarrow$ buy that side, **HOLD TO RESOLUTION** (don't try to flip — exit liquidity is poor).

Worked example: Pinnacle implies Lakers 70% (de-vigged). Polymarket "Lakers win" = **62¢**. Edge = $70 - 62 = +8$ points $\rightarrow$ buy at 62¢, hold. If your prob source is right, this is +EV; over hundreds of such bets you profit.

C3. De-vig (strip the bookmaker margin)

Raw book odds include a margin ("vig") so implied probs sum to $>100\%$. Remove it:

- **Multiplicative** (default, 2-way): each side's implied prob $\div$ sum of all implied probs. e.g. 0.55 & 0.50 sum $1.05 \rightarrow 52.4\%$ & 47.6% .
- **Power**: exponent-based; handles favorite/longshot asymmetry; between multiplicative and Shin.
- **Shin** (most theoretically sound): assumes part of the vig is insider info; allocates more margin to longshots; iterative; **best for heavy favorites, player props, multi-way markets**.
- **Rule**: multiplicative for simple 2-way; Shin/power for heavy-favorite or multi-outcome. Free devig calculators: OddsJam, gamblingcalc, edgeslip.

C4. Which markets (and the threshold)

- Polymarket sports: **NFL, NBA, MLB, NHL, soccer, UFC, boxing, tennis, F1, Olympics, World Cup**; markets = game winners, season/championship, MVP, some props.
- **Edge threshold**: $\geq 5\%$ commonly cited; you need **$\sim 2.2\text{--}3.0\%$ just to cover** the 2% winner fee + gas + slippage, so demand more than that.
- Best opportunities: **less-liquid / less-watched markets** where Polymarket lags **minutes–hours** after news (injuries, lineups). Avoid racing bots on the obvious 15–60s edges.

C5. Closing Line Value (CLV) — your proof of edge

The **closing line** (final odds before kickoff) bakes in *all* info → the gold-standard "true probability." **If you consistently get a better price than the closing line, you are mathematically profitable long-term** (Law of Large Numbers) — *before* results even confirm it.

- **Beat it by:** (1) **line-shop** the best price, (2) **bet EARLY** (soft lines before sharp money moves them), (3) make genuine +EV bets.

- **Track:** log your entry price + the closing line for 100+ bets; compute % positive-CLV and average CLV.

- **Benchmarks:** any positive CLV = beating the market; **+5% avg CLV = excellent; 60–65% of bets beating CLV over 200+ = a confirmed, durable edge.** *This is how you KNOW you have an edge without waiting months for variance to resolve.*

C6. Sizing & tools

- **Kelly:** $f = (b \cdot p - q) / b$; use **Quarter Kelly**; $\leq 5\%$ bankroll/bet; flat tiny stakes while learning.
- **Tools:** **OddsJam** (edge feed + devig + CLV + prediction-market trader tracker), **PredictEngine** (arb/value scanner), **Pinnacle Odds Dropper**, odds APIs (oddspapi). Trackers: polysight.app, polytraders.io. Repos: Alirun/polymarket-trader, jon-becker/prediction-market-analysis (36GB Polymarket+Kalshi history for **backtesting**).

C7. Step-by-step workflow

1. Pick a sport/league you understand. 2. Pull Pinnacle (or Betfair) lines. 3. De-vig → true prob. 4. Scan Polymarket for the same games; find price < true prob by $\geq 5\%$ (after costs). 5. Place a **limit** order (0 fee) and **hold to resolution**. 6. **Log entry price + closing line** → track CLV. 7. After 100–200 bets, check CLV%: beating it 60%+ → real edge, scale slowly with Quarter Kelly. Not beating it → fix your inputs before risking more.

C8. Flaws & mitigations (honest)

- **Bots own the fast/big edges** (15–60s) → don't race them; value-bet slower markets and hold.
 - **Liquidity:** fewer markets than a sportsbook; low-liquidity markets are easy to **buy, hard to sell** → hold to resolution, size to available depth.
 - **Wash trading** inflates Polymarket volume (Columbia study) → don't trust volume blindly; check real two-sided depth.
 - **Rising efficiency** = fewer easy edges over time; **costs** (2% fee + spread) must be cleared; value betting is the **highest skill bar** (~3% of accounts skilled).
 - **Access:** Polymarket blocked in AU (same gate as weather).
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PART D — WEATHER vs SPORTS: WHICH, WHEN, HOW TOGETHER

Weather	Sports value-betting
Five-ensemble forecasts are the key	De-vigged Pinnacle ($r^2=0.997$)
Logic (objective) strength	Strongest (near-unfalsifiable)
Best for automating; objective resolution	Year-round; if you know sport
Fast (fast learning) speed	Per-game
Minor edge; bots in big cities have a flaw	Liquidity/exit; bots on fast edges
More secondary cities, limit orders	Less-liquid markets, hold to resolution
Proved win% after costs is the metric	CLV (beat closing line 60%+)
Automatable? (tokens template)	Partly (scanners + manual judgment)
Edge size (model vs price)	$\geq 5\%$ (true prob vs price)

Which to start with?

- Want **objective, automatable, fast feedback** → **Weather** (and you have Claude Code to build the bot).
- **Know a sport deeply** and want the **most bulletproof logic** + year-round volume → **Sports**.
- Ideal: learn **both** — they share the one skill and diversify your edge across unrelated domains.

The validation gate (same for both):

1. **Paper-trade** (fake money) for weeks → prove you can spot edges. 2. **Tiny real money** (\$1–5/bet, few-hundred-\$ bankroll), **log everything** (sports: track CLV). 3. **100+ trades** → positive after costs / beating CLV 60%+? → scale slowly (Quarter Kelly, $\leq 5\%$ /bet). 4. Only then **automate / size up**. Each phase must succeed before the next. **Never skip to big money.**

PART E — QUICK REFERENCE

Formulas: $EV = p \cdot (\text{win}) - (1-p) \cdot (\text{loss})$. Kelly $f = (b \cdot p - q)/b$, use $0.25 \times$. De-vig (mult.) = $\text{implied_prob} \div \Sigma \text{implied_probs}$.

Edge thresholds: Weather ≥ 8 pts; Sports $\geq 5\%$ (clear $\sim 2.5\%$ costs first).

CLV benchmarks: $+5\%$ avg = excellent; 60–65% of bets beating close over 200+ = durable edge.

Weather data: Open-Meteo (ensembles), api.weather.gov (NWS + METAR), Windy (compare models), Tropical Tidbits. Runs: GFS 00/06/12/18 UTC, ECMWF 00/12 UTC. Stations: KLGA (NYC), KDAL (Dallas). Secondary cities: Buenos Aires, Cape Town, Dallas, Atlanta.

Sports data: Pinnacle / Betfair / PS3838 (sharp lines); OddsJam, PredictEngine, Pinnacle Odds Dropper, oddspapi (APIs); devig calcs ([oddsjam/gamblingcalc/edgeslip](https://oddsjam.com/gamblingcalc/edgeslip)).

Repos: [natestokens/polymarket-weather-bot](https://github.com/natestokens/polymarket-weather-bot), [alteregoeth-ai/weatherbot](https://github.com/alteregoeth-ai/weatherbot), [Polymarket/polymarket-cli](https://github.com/Polymarket/polymarket-cli), [Polymarket/py-clob-client](https://github.com/Polymarket/py-clob-client), [Alirun/polymarket-trader](https://github.com/Alirun/polymarket-trader), [jon-becker/prediction-market-analysis](https://github.com/jon-becker/prediction-market-analysis) (backtest data).

Rules of survival: bet small ($\leq 5\%$), limit orders (0 fee), hold to resolution, log everything, validate 100+ trades, never bet what you can't lose, never all-in.

Access gate: Polymarket + Kalshi block Australia — solve venue/legality before funding.

Scam watch: never run commands like `powershell irm <site> | iex` ("install a bot" = malware); guard your wallet private key; assume "\$X/day bot" promos are paid affiliate funnels.

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